

Ashwin Subramanian Murugan

Email: ae22b001@smail.iitm.ac.in | Phone: +91 6382074424

EDUCATION

Indian Institute of Technology (IIT), Madras

B.Tech in Aerospace Engineering

Department Rank: 1/75

Expected June 2026

CGPA: **9.92/10**

TEST SCORES

- Secured **All India Rank 156** in JEE Advanced 2022 out of **150K+** candidates Aug 2022
- Secured **All India Rank 122** in JEE Mains 2022 out of **1M+** candidates Jul 2022
- Secured **All India Rank 239** in Kishore Vaigyanik Protsahan Yojana (KVPY SX) 2021 and awarded the KVPY Fellowship by Department of Science and Technology, Govt. of India Jan 2021

HONOURS AND AWARDS

- Awarded **Sri S Subramanian Prize** for securing the **highest CGPA** among **1500+** students in the first year of the B.Tech program at IIT Madras Apr 2024
- Awarded **Prof T K Varadan Prize** for achieving the best academic performance in the second year of the B.Tech Aerospace Engineering program at IIT Madras Apr 2025
- Secured **Runner-Up** in the **James Dyson Award 2024** at the National level for developing **HydroChurn** - a portable, self-sustaining water bottle Sept 2024
- Received the United Nations Millennium Fellowship 2025 to further develop **HydroChurn** Aug 2025
- Secured First Runner-Up in the **Airbus Flight Challenge 2024** at Shaastra, IIT Madras for developing a model to predict the dynamic effects of sudden braking on an airplane Jan 2024
- Received the Indian Academy of Sciences Summer Research Fellowship 2024 for a fully funded summer research internship at IISC Bangalore May 2024

PUBLICATIONS

Ashwin Subramanian Murugan, T. M. Muruganandam. "Shock Train Stability in Scramjet Isolators under Back-Pressure Fluctuations." *Manuscript in preparation* for submission to the *Journal of Fluid Mechanics*, December 2025.

J. Hwang, **Ashwin Subramanian Murugan**, Q. Angéloz. "Characterization of Helmholtz Resonators for Acoustic Force Generation and Microflight." *Manuscript in preparation* for submission to *Science Advances*, December 2025.

RESEARCH AND INDUSTRY EXPERIENCE

École Polytechnique Fédérale de Lausanne (EPFL)

May 2025 – Jul 2025

Project: Development of Acoustically Actuated Microflyers

Lausanne, Switzerland

Advisor: [Prof. Selman Sakar](#)

- Designed and analyzed one of the smallest artificial flying robots, a millimetre-scale acoustically actuated microflyer using Helmholtz resonators for controlled flight as a part of the E3 Summer Research Program (2% acceptance rate)
- Fabricated microflyers with integrated propellers and resonators using a Nanoscribe 2-photon polymerization system
- Performed experiments with a 40 kHz transducer array, demonstrating lift generation and controllable flight using acoustic power

Indian Institute of Technology Madras

Jul 2023 – Present

Project: Effects of Back Pressure Fluctuation on Pseudo-Shock Waves in a Scramjet Isolator

Chennai, India

Advisor: [Prof. T M Muruganandam](#)

- Investigating the effect of back pressure fluctuations on shock train stability within a scramjet isolator across varying flow conditions
- Performing high-fidelity Large Eddy Simulations (LES) using OpenSBLI to resolve unsteady shock-boundary layer interactions and pressure wave dynamics
- Working on determining isolator design criteria to prevent unstart under fluctuating back-pressure conditions

Agnikul Cosmos

Nov 2024 – Jan 2025

Propulsion Engineering Intern

Chennai, India

Agnikul Cosmos is an IIT Madras-based aerospace startup creating 3D-printed, semi-cryogenic launch vehicles for affordable small satellite launches

- Modeled and analyzed the **LOX subcooling, pumping, and filling systems** for the booster stage testing facility using AFT Fathom, determining flow rates, pressure distributions, and valve operation parameters
- Simulated transient thermal behaviour of the LOX system using **NASA's GFSSP** to evaluate critical parameters related to the liquid nitrogen subcooling system, the liquid oxygen reservoir, and the booster stage filling system
- Analyzed **heat transfer and thermal stratification** in the booster stage propellant tanks to determine propellant requirements

Indian Institute of Science Bangalore

May 2024 – Jul 2024

Project: Numerical and Experimental Study on Shock Wave Pulsations over Spiked Cylinder

Bangalore, India

Advisor: [Prof. Duvvuri Subrahmanyam](#)

- Conducted experiments in the **Roddam Narasimha Hypersonic Wind Tunnel (RNHWT)** to investigate shock wave pulsations at **Mach 6**
- Performed hypersonic flow simulations using Ansys Fluent to analyze the dynamics of shock wave pulsations over a spiked cylinder and validated it using experimental data
- Performed a literature review to understand the driving mechanisms of shock wave pulsations

Indian Institute of Technology Madras

Jul 2025 – Present

Project: Deposition and Erosion in Solid Rocket Motor Nozzles

Chennai, India

Advisor: [Prof. P. A. Ramakrishna](#)

- Performing multiphase CFD simulations of solid rocket motor (SRM) nozzles to analyze particle-induced erosion and deposition mechanisms using empirical modeling approaches
- Carrying out full-scale SRM firing experiments to provide quantitative benchmarks for erosion and deposition model validation

TECHNICAL PROJECTS

Course Project - Data-Driven Modeling of Aerospace Systems

Jul 2025 – Nov 2025

IIT Madras

Chennai, India

- Developed a Fourier Neural Operator-based model for the prediction of transient flows over parameterized elliptical geometries
- Designed a Reduced Order Model-based controller for morphing-airfoil actuation, enabling constant-lift performance under highly turbulent flow conditions

Student Unmanned Aerial Systems Competition (SUAS) 2025

Nov 2024 – Present

Team Advisor: [Prof. Satadal Ghosh](#), IIT Madras

Chennai, India

- Part of a 15-member team developing an autonomous quadcopter UAV for SUAS 2025, focusing on autonomous flight, real-time obstacle avoidance, object detection, and precision payload delivery
- Implemented computer vision algorithms to detect, classify, and geolocate ground markers from altitudes of up to 75 feet, utilizing synthetic datasets generated in Blender
- Executed autonomous mission laps at SUAS 2025, integrating waypoint navigation, real-time object detection, and autonomous mapping of specified mission areas

- Currently collaborating with commercial partners to adapt and deploy this technology for domain-specific applications

AstroSim

Physics and Astronomy Club, Centre for Innovation (CFI), IIT Madras

May 2023 – Mar 2024
Chennai, India

- Developed a C++ program to numerically solve the N-body problem, investigating complex system behaviours and stellar dynamics
 - Applied advanced numerical methods and optimization techniques, including the Barnes-Hut Octree algorithm, 3-D inelastic collisions, and parallel processing, achieving an **8x** performance improvement
 - Simulated the Andromeda - Milky Way collision with over **100K+** objects, demonstrating the scalability of the system for large-scale astrophysical simulations
- Link: [Github Repository](#)

TECHNICAL SKILLS AND COURSEWORK

Programming and Software :	Python, MATLAB, C++, Ansys Fluent, L ^A T _E X, Solidworks, Comsol, AFT Fathom, NASA's GFSSP, AutoCAD, Fusion 360
General Purpose Tools :	Microsoft Office, Blender, Canva, Adobe Illustrator
Coursework :	· Rocket Propulsion, Gas Turbines, Aerodynamics, Gas Dynamics, Vibrations, Fluid Mechanics, Thermodynamics, Flight Dynamics, Gas Dynamics Lab, Aerospace Structural Mechanics · Computational Fluid Dynamics, Scientific Computing · Linear Algebra, Multivariable Calculus, Complex Analysis · Econometrics, Principles of Economics, Basic Electrical Engineering

CO-CURRICULAR ACTIVITIES

Coordinator

Coding and Logic Vertical, Shaastra, IIT Madras

May 2023 – Jan 2024

- Coordinated a nationwide competitive programming contest for college students, managing both the preliminary and final rounds to test advanced programming and analytical skills
- Developed and curated challenging programming problems for IIT Madras' first rated CodeChef contest, attracting over **25K+** participants worldwide, and coordinated the finals held at IIT Madras

Freshman Mentor

Saathi – Student Wellness Cell, IIT Madras

Jul 2024 – Apr 2025

- Mentored four first-year students by providing guidance on academic and personal development, supporting their transition to college, and assisting with coursework and campus resources

EXTRA-CURRICULAR ACTIVITIES

Astronomy Enthusiast

- Was a member of St. John's Public School's Astronomy Club, where I participated in hands-on STEM activities and presented weekly updates on space science
- Conducted night-sky observations using telescopes, including planetary imaging, as part of Horizon - the Physics and Astronomy Club at IIT Madras

Keyboardist and Violinist

- Completed Trinity Grade 2 in Keyboard and Grade 2 in Music Theory
- Amateur violinist specializing in Indian classical music
- Was a member of the school choir and performed multiple times in the school assembly and Annual Day